



Financial risk management

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Recent developments in financial risk management

Introduction

Financial risk management (FRM) has been a very hot topic in recent years and updates are frequent. When it comes to financial risk management, the activities of the Basel Committee on Banking Supervision (the Basel Committee), the primary global standard setter for the prudential regulation of bank, and national financial service regulators often come to mind. However, financial risk management is much broader in terms of its spectrum and variety.

Wide application and spectrum of FRM

FRM is not just part of the work of many who work in the financial services industry or corporate treasury departments; it in fact has strong interplay with many professional standards and disciplines.

Take accounting standards as an example. Accounting standards are becoming more aligned and synchronized with FRM. For example, Hong Kong Financial Reporting Standards (HKFRS) 13 *Fair Value Measurement* incorporates relevant requirements on credit value adjustment and debit value adjustment. HKFRS 9 *Financial Instruments* adopts a new forward-looking approach in relation to credit risk provisioning compared to the previous standard which included an incurred loss concept.

In particular, HKFRS 9 is an important accounting standard for FRM as it relates to credit risk provisioning (i.e. expected credit loss). This new standard requires more modeling and risk management techniques in the calculation of expected credit loss, for instance, the estimation of life-time of certain non-contractual type of products through behaviouralisation technique and probability of default modeling.

Professional accounting bodies are also revising their education curricula. For example, the Hong Kong Institute of Certified Public Accountants is introducing its New Qualification Programme, to tie in with the latest industry standards. The curriculum is also more diversified and extensive in its content, covering various up-to-date financial risk management topics, and the soft skills and ethical foundations needed for career success.

In short, the area of FRM has seen a lot of evolutions over recent times. With more and more technical jargon or terms which look quite alike and can be hard to differentiate, it may be worthy going through a few of them below together.

Key concept analysis

Among various key financial risks, credit risk is one of the important risks that is faced by many organizations almost every day. In fact this term is never unfamiliar.

Firstly, there are many terms related to credit risk which are regularly included in media reports, like changes of credit rating of a corporate and depreciation of collateral. Secondly, many people in Hong Kong usually have more than one credit card and in fact when we are applying for a credit card, the limit we are offered is correlated with our credit risk evaluated by the financial institution through credit risk assessments.

So, while we regularly see credit risk related terms, they, however, are not as straightforward as they first appear.

1. Credit risk

Traditionally, credit risk is unilateral (i.e. one-sided) in that the lender provides funding to the borrower and the lender faces credit risk due to the fact that the borrower may default on payment.

However, are all credit risks unilateral?

Counterparty credit risk (CCR) is a type of credit risk that is very different given its bilateral nature. The counterparty to a transaction could default before the final settlement of the transaction in cases where there is a bilateral risk of loss.

A typical bilateral nature of transaction can be derivative transaction and, if we take a simple fixed-to-float interest rate swap as an example, the mark-to-market gain/loss can vary depending on the movement of underlying factors. Sometimes the counterparty is at a gain position but he or she can also be at a loss position before settlement. With this bilateral nature of payoff outcome, potential credit risk can happen to both parties of the derivative transaction.

In addition, credit risk does not necessarily purely look at default risk. For example, there can be risks associated with the deterioration in the creditworthiness of the counterparty to a transaction and the credit spread risk.

The financial services industry often faces credit risks, particularly for those performing financial intermediary roles. One of the key standards - Basel III – is an internationally agreed set of measures developed by the Basel Committee in response to the global financial crisis of 2007-09 ^{note 1}. These measures aim to strengthen regulation, supervision and risk management of banks. The risk-based capital charges for CCR in Basel III cover two important characteristics: the risk of counterparty default and a credit valuation adjustment (CVA).^{note 2}

2. Credit rating

In the media, we often read about credit risks such as changes in the credit rating of both corporates and countries, issued by various rating agencies including but not limited to Moody's Investors Service, S&P Global Ratings, Fitch Ratings, and local rating agencies in certain jurisdictions.

However, credit rating is more complex than we think.

If we take a closer look at commercial credit rating, this can be further divided into internal and external ratings. External ratings are those we usually hear from the media and are the external rating of a counterparty issued by rating agencies. Nonetheless, some organizations also devise their own internal rating of their counterparty based on their own credit data and a credit master-scale.

Apart from internal and external ratings, there are also differences between issue and issuer rating. Issue rating concerns the creditworthiness of an obligor with respect to a specific financial obligation, a specific class of financial obligations, or a specific financial programme; whereas issuer rating is normally an opinion about an obligor's overall creditworthiness.

One of the differences between issuer and issue rating is that issue rating considers the creditworthiness of guarantors, insurers, or other forms of credit enhancement on the obligation while also taking into account the currency in which the obligation is denominated.

3. Collateral

It is quite common for a lending transaction to have collateral requirements. However, unlike some straightforward cases that are usually quoted in traditional educational material, there is often currency mismatch between the collateral and the lending itself.

In addition, collateral comes in multiple forms. Some collateral can be more liquid and can be sold more easily when needed, for example shares in publically listed companies; others are the opposite, e.g. capital equipment or unlisted shares. There are different FRM measures or techniques in handling various scenarios including the application of different haircut rates, i.e. the difference in the value of collateral in the market and as ascribed when used as collateral.

On top of placing collateral, guarantee is another form of credit risk protection technique and receives a different set of FRM treatments.

4. Modeling

Apart from the definitions of various key terms in the FRM spectrum, financial risk measurements and modeling requirements have also seen constant changes to tie in with modern financial risk management practices and the latest professional standards and requirements. For example, value-at-risk (VAR) has long been a common risk measurement, but this is being superseded by expected shortfall modelling. There have also been increased considerations of both through-the-cycle and point-in-time probability of default for expected credit loss calculation, due to the implementation of HKFRS 9.

Key takeaways

Credit risk is just one of the many financial risks and each of the financial risk has its own characteristics and uniqueness. However, given the broadness and the complexity of various FRM topics, we cannot go through each and every key topic in this article with sufficient coverage. Think of this article as a starting point for your further research into the topics.

It is clear that that FRM covers a very wide spectrum including but not limited to risk classification, risk measurement, risk monitoring and risk mitigation like hedging.

FRM functions of an organization are likely to become even more challenging, not only due to its high connectivities, but also arrival of new regulations (e.g. Basel IV). Financial risk managers need to equip themselves with the technical expertise and soft skills necessary to quickly embrace the developments, and respond to their employers' and clients' needs.

Footnotes

1. <https://www.bis.org/bcbs/basel3.htm>
2. Counterparty credit risk in Basel III - Executive Summary
https://www.bis.org/fsi/fsisummaries/ccr_in_b3.htm

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